Engineer to Excel

SIMATS Engineering

Microservices Architecture

in Cloud Environments

Technical Presentation
CSA1513- CLOUD COMPUTING AND BIG DATA ANALYTICS

Presented by: G VENKATA GOUTHAM REDDY
REG NO: 192324265

1. Objective

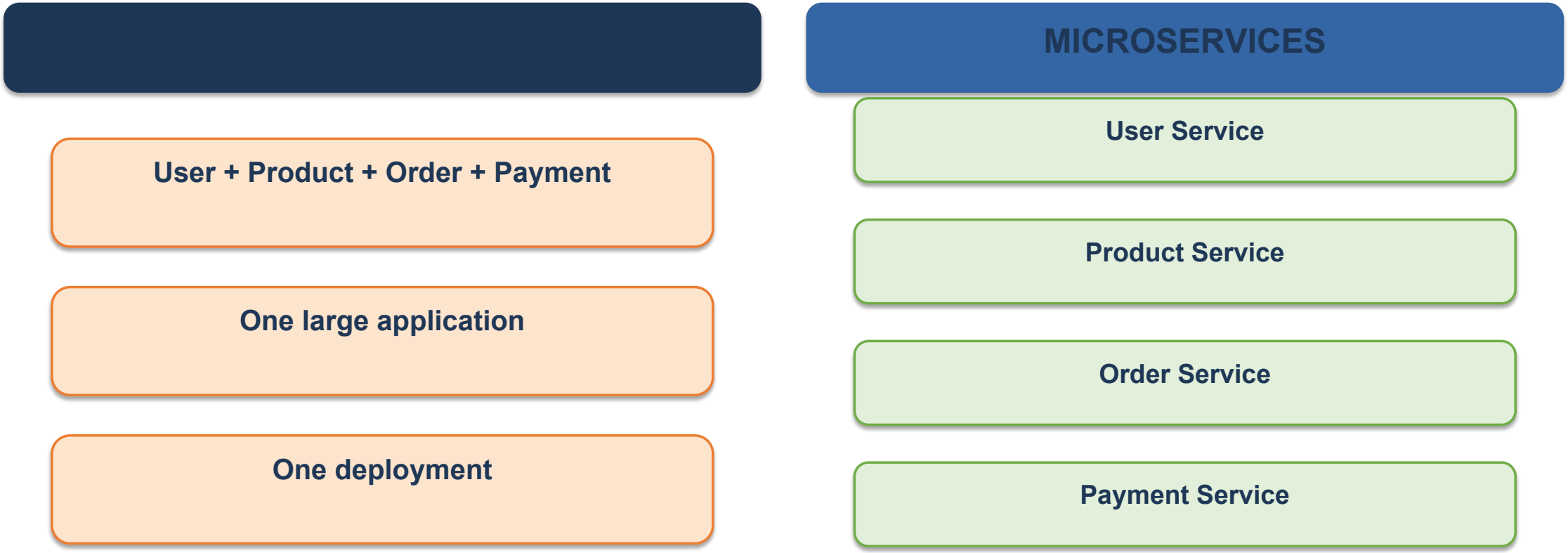
- Understand what microservices are.
- Compare microservices with monolithic architecture.
- Understand how microservices work in the cloud.
- Learn the main benefits and challenges.
- See one real-world example.

2. What are Microservices?

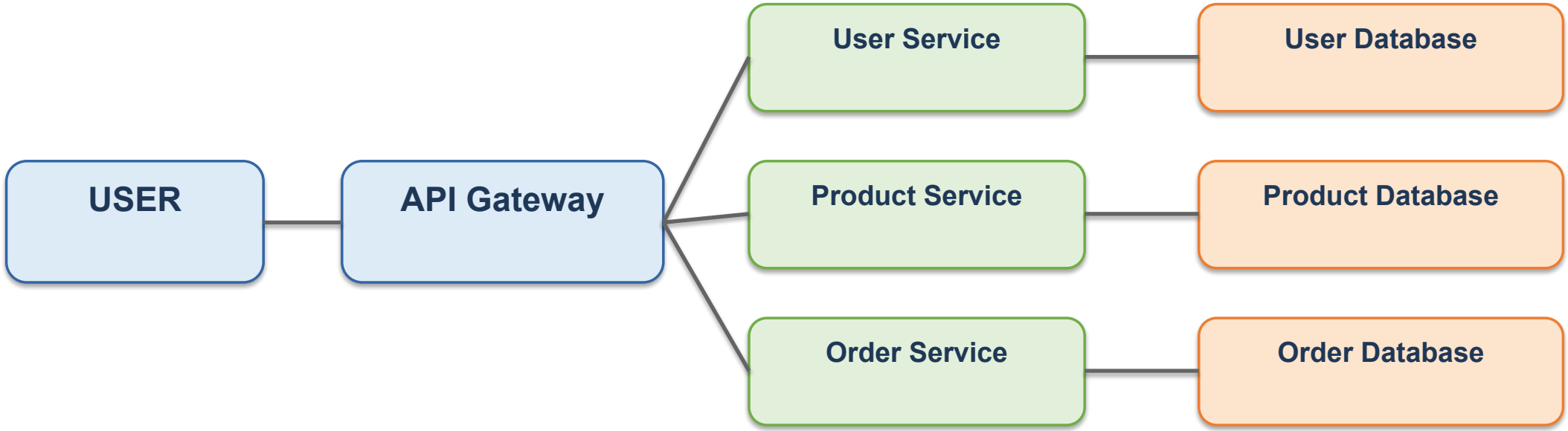
Microservices divide one large application into small, independent services.

- Each service performs one specific job.
- Services communicate using APIs.
- Each service can be updated separately.
- Each service can be scaled when needed.

3. Monolithic vs Microservices



4. Simple Microservices Architecture





Simple flow: User → API Gateway → Service → Database

5. Benefits in Cloud

Easy Scaling

Scale only the busy service.

Independent Updates

Update one service without changing everything.

Fault Isolation

One service failure may not stop the whole app.

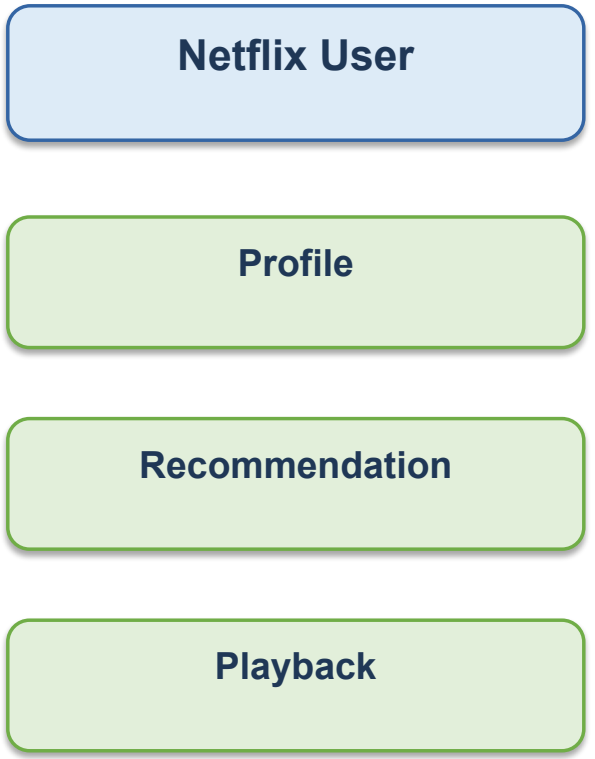
Faster Development

Different teams can work on different services.

6. Real-World Example: Netflix



- Netflix uses many independent services.
- Different services handle different functions.
- Examples: user profiles, recommendations, search and video playback.
- Cloud infrastructure allows services to scale for large numbers of users.



7. Challenges

- More services mean more system complexity.
- Services depend on network communication.
- Monitoring and debugging can be harder.
- Security must be handled between services.
- Microservices may be unnecessary for small applications.

Use microservices when the application's scale or complexity justifies them.

8. Conclusion

- Microservices divide a large application into smaller services.
- Each service can work and scale independently.
- Cloud platforms make deployment and scaling easier.
- Microservices improve flexibility, but they also add complexity.
- Netflix is an example of microservices used at large scale.

Key Takeaway: Small services + Cloud = Scalable and Flexible Applications